Saumik Dana

https://saumikdana.github.io/

Summary

US green card holder with PhD from The University of Texas at Austin and experience in feature engineering, optimization, stochastic modeling, and LLM pipelines. Comfortable owning research and infrastructure in end–to–end production systems.

Experience

Quantitative Researcher, **Zebra Capital Management LLC**, Stamford CT, **Feb 2024 - Dec 2025**

ML — backed systems

- Built intraday options trading strategies driven by Bayesian anomaly–detection signals
  - Implemented nonlinear optimization and FFT options pricing based stochastic volatility model calibrations
  - Generated *weekly returns of* $5 - 10\%$ *on peak capital at risk*
- Deployed live equity portfolio with NSGA-II genetic-algorithm optimization and regime-adaptive rebalancing
  - Calibrated Hidden Markov Model backed regime detection and Jensen – Shannon distance based rebalancing
  - Set up optimizer workflow: per-ticker α/β predictions → Bi-objective function → Pareto-optimal weights
  - Achieved *5+% annualized alpha-to-benchmark on multi-year backtest*
- Deployed multi-asset interday options trading strategies using pre-trained models
  - Trained TabPFN classifiers with backtested PnL as target and LoRA-adapted Amazon Chronos based rolling embeddings of technical indicators as features

LLM — backed Agents

- Built MCP–wrapped tool–calling backed trade signal mining pipeline – Discovered *1.5+ Sharpe strategies*
- Built vision backed (*qwen3p7-plus*) PDF QA system with contrastive-embedding fine-tuning and Qdrant indexing

LLM — backed systems

- Implemented intraday options trading driven by reasoning (*openai/gpt-oss-120b*) over stochastic volatility signals
  - Designed state-dependent prompt architecture with verdict history and entry-thesis read-back
  - Engineered feature pipeline distilling versioned prompts evaluated over fixed windows — prompt is the strategy
- Deployed news and price action narrative fed intraday FX trading system
  - Built staged pipeline with explicit CoT scaffolding sized to a small non-reasoning model (*llama-3.1-8b-instant*)
  - Closed the loop with DSPy (GEPA) prompt optimization: archived live–runs with price action as rewards
- Deployed news and sovereign yield-curve differentials narrative fed intraday commodity ticker trading system
  - Engineered reasoning (*openai/gpt-oss-120b*) pipeline deriving daily signal from evidence — attribution, conflict–resolution, abstention rules

Infrastructure

- Designed serverless stacks on AWS (Lambda, EventBridge, DynamoDB, S3, CloudFormation) and Modal
- Engineered GitHub Actions CI/CD pushing digest-pinned ECR images shared across multiple bots
- Handed GitHub secrets to bot-specific YAML templates with DST-aware EventBridge scheduling
- Tuned Lambda for ML inference with multi-GB ephemeral storage and S3-backed HuggingFace/Torch caches
- Built React/static dashboards over DynamoDB/S3, served via FastAPI behind Lambda Function URLs

Computational Engineer, **VISIE Inc.**, Austin TX, **Aug 2023 - Nov 2023**

- Joined during early integration of imaging and robotic actuation in surgical navigation platform
- Implemented TCP/UDP communication protocols to control robotic arm motion
- Supported product packaging, deployment, live demonstration used in *successful Series A round*

Computational Lead, **Sophelio**, Austin TX, **Aug 2022 - Mar 2023**

- Adapted physics-informed modeling—originally for fusion experiment data—to financial time series
- Engineered production pipeline: sparse regression for PDE construction, signal generation
- Implemented CAGR maximizing Bayesian TPE optimizer based swing trading system

Education

Doctor of Philosophy in Engineering Mechanics — **University of Texas at Austin**, Austin TX
Master of Engineering in Mechanical Engineering — **Indian Institute of Science**, Bangalore, India

Bachelor of Engineering in Mechanical Engineering — University of Mumbai, Mumbai, India

Technical Skills

Languages/Data — Python, pandas, Polars, NumPy, requests, BeautifulSoup, RSS/XML
Cloud — AWS (Lambda, DynamoDB, S3, ECR, EventBridge, API Gateway, CloudFormation), Modal, Docker
LLM — Groq, Langchain, Fireworks AI, Qdrant, DSPy, MCP, Tool calling
APIs — FastAPI, React/JSX, Next.js, HTML, Alpaca, OANDA, TradeStation, yfinance
ML/Quant — PyTorch, scikit-learn, TabPFN, Chronos, PEFT, hmmlearn, SciPy, pymoo, autograd